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Solar ESS product sheet · October 2026

Breakers

CHINT MCCB, ABB and IPG

Sized to the cable and the inverter on every circuit, DC and AC.

> **A breaker is cheap until the day it has to work. This is the wrong place to save money.**

What we fit

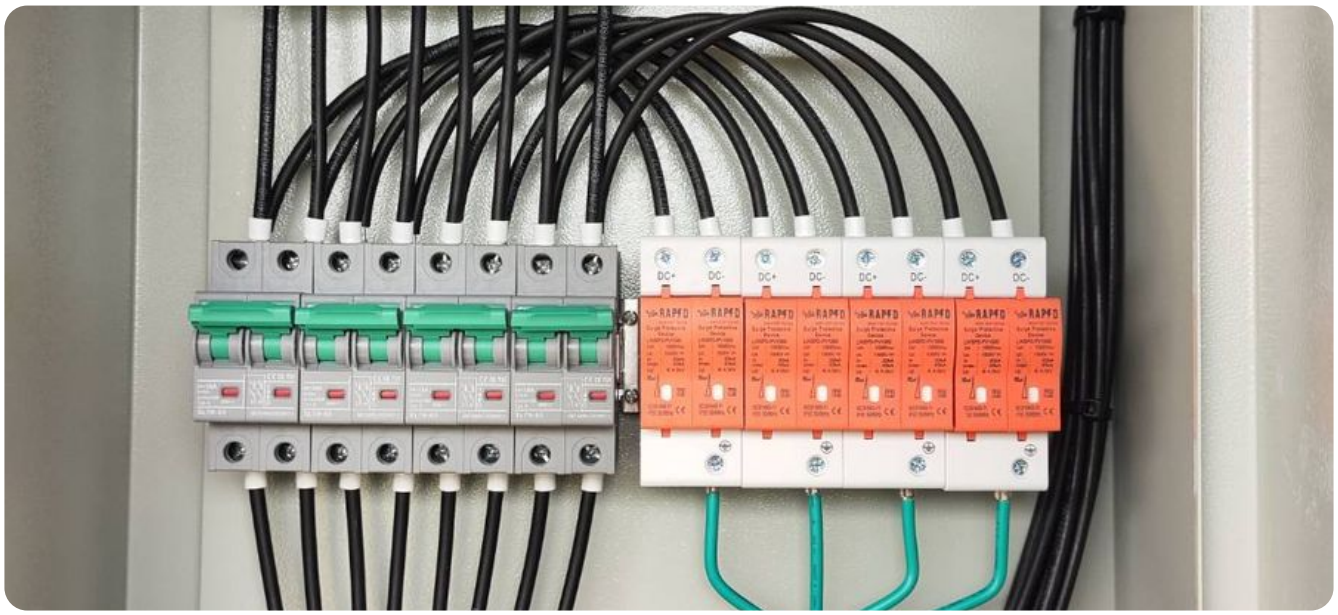
AC breakers	ABB, IPG
DC breakers	SUNTREE
DC fuse holders	SUNTREE
MCCB, main and changeover boards	CHINT

Why we pick it

- Each breaker is matched to the cable it protects, so the breaker trips before the cable overheats.
- DC circuits get breakers rated for DC. An AC breaker on a DC string cannot be relied on to break the current.
- Labelled and laid out in the box so the next electrician can see what each one does.

DC combiner box: SUNTREE fuse holders and breakers

Every solar string comes into the DC box through a SUNTREE fuse holder on both the positive and negative lines, then a two-pole SUNTREE DC breaker, before it reaches the inverter.



DC combiner box: SUNTREE fuse holders and two-pole DC breakers, one set per string

- The fuse protects the string cable and panels if another string back-feeds a fault into it.
- The two-pole DC breaker isolates each string on its own, so one string can be worked on while the rest keep producing.
- Fuse holders open safely for checking and changing a fuse, with no live parts exposed.

Moulded-case breakers (MCCB)

On the main feeders of a hybrid or backup system (the grid, inverter, backup and bypass circuits) we fit CHINT moulded-case circuit breakers. They carry the full feeder current, from about 63 A to 160 A and above, where small DIN-rail breakers stop.



Changeover board: grid, inverter, backup and bypass MCCBs with a 4-pole ATS

- A high breaking capacity, so a short circuit is cleared safely instead of welding the contacts shut.

- Earth-leakage versions on the inverter and backup circuits, for protection against electric shock as well as overload.
- Four-pole models where the neutral must switch with the phases, as on a changeover board.
- Every breaker is labelled, and a bypass breaker with written switching steps keeps the building supplied while the ATS or inverter is serviced.

RCBO: one-way or two-way?

An RCBO combines earth-leakage, overload and short-circuit protection in one unit. In a solar home, the direction power flows through it matters.

Unidirectional	Built for power in one direction only, from the supply (Line / In) to the circuit (Load / Out). For ordinary lighting and socket circuits fed only from the grid.
Bidirectional	Certified by the maker to work safely with power flowing either way. For any circuit that solar, a battery, a generator or a V2G charger can feed.

How to tell them apart

- A unidirectional RCBO has terminals marked Line / Load or In / Out, or an arrow, on the case or the front diagram.
- A bidirectional RCBO has no fixed supply side, or its datasheet states that it is suitable for power flowing in both directions.
- If the datasheet does not say bidirectional, we treat it as unidirectional.

How we wire them

- A unidirectional RCBO only ever takes the supply on its Line / In terminals. An inverter, battery or generator is never connected to its Load side; fed backwards it may not trip, can be damaged, and can leave its electronics live during a fault.
- On circuits that solar, a battery or a generator can feed, we fit RCBOs the maker certifies as bidirectional, wired to the maker's diagram.
- Where the neutral must be switched, we use double-pole devices.



CHINT MCCBs labelled GRID-INV and GRID

For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1. All qualifications and certificates will be furnished upon request or upon completion of contract.